📘 **AI/ML Internship – Day 17 Notes & Tasks**

🟢 **Module 3 : Useful Python Libraries for AI**

📌 **Part 1: Why Libraries are Important?**

🔹 **What is a Library?**

A library is a collection of pre-written code that helps you perform tasks easily.

🔹 **Why Use Libraries?**

- Saves time
- Reduces code complexity
- Used in real-world AI projects

👉 Example:

- Instead of writing math manually → use library

📌 **Part 2: Math Module (Basic AI Calculations)**

🔹 **Import**

```
import math
```

🔹 **Common Functions**

```
print(math.sqrt(16))     # Square root
print(math.pow(2, 3))    # Power
print(math.ceil(2.3))    # Round up
print(math.floor(2.9))   # Round down
```

🔹 **Constants**

```
print(math.pi)
print(math.e)
```

📌 Part 3: Random Module (Data Simulation)

🔹 Import

```
import random
```

🔹 Examples

```
print(random.randint(1, 10))
print(random.choice([10, 20, 30]))
print(random.random())
```

👉 Used in:

- Testing
- Simulation
- ML experiments

📌 Part 4: Datetime Module (Time-Based Data)

🔹 Import

```
import datetime
```

🔹 Current Date & Time

```
now = datetime.datetime.now()
print(now)
```

🔹 Format Date

```
print(now.strftime("%Y-%m-%d"))
```

👉 Used in:

- Logs
- Time-series data

- Real-world datasets

📌 **Part 5: Introduction to Data Visualization**

👉 Before ML, we need to **see data**

🔹 **What is Visualization?**

Displaying data using graphs.

🔹 **Install Library**

```
pip install matplotlib
```

🔹 **Basic Plot**

```
import matplotlib.pyplot as plt

x = [1, 2, 3]
y = [10, 20, 30]

plt.plot(x, y)
plt.show()
```

🔹 **Bar Chart**

```
plt.bar(x, y)
plt.show()
```

👉 Used in:

- Data analysis
- ML insights
- Reporting

📌 **Part 6: Real-World Thinking**

👉 In AI projects:

- math → calculations
- random → simulations
- datetime → time data
- matplotlib → visualization

🎯 **Tasks for Day 17**

🧠 **Task 1: Theory**

1. What is a Python library?
2. Why libraries are important?
3. What is math module?
4. What is random module used for?
5. What is data visualization?

💻 **Task 2: Math Practice**

1. Find square root
2. Find power
3. Round numbers
4. Use π value

💻 **Task 3: Random Practice**

1. Generate random number
2. Pick random element
3. Generate list of random numbers

💻 **Task 4: Datetime Practice**

1. Print current date
2. Format date
3. Extract year, month

💻 **Task 5: Visualization**

1. Create line graph
2. Create bar chart
3. Plot student marks